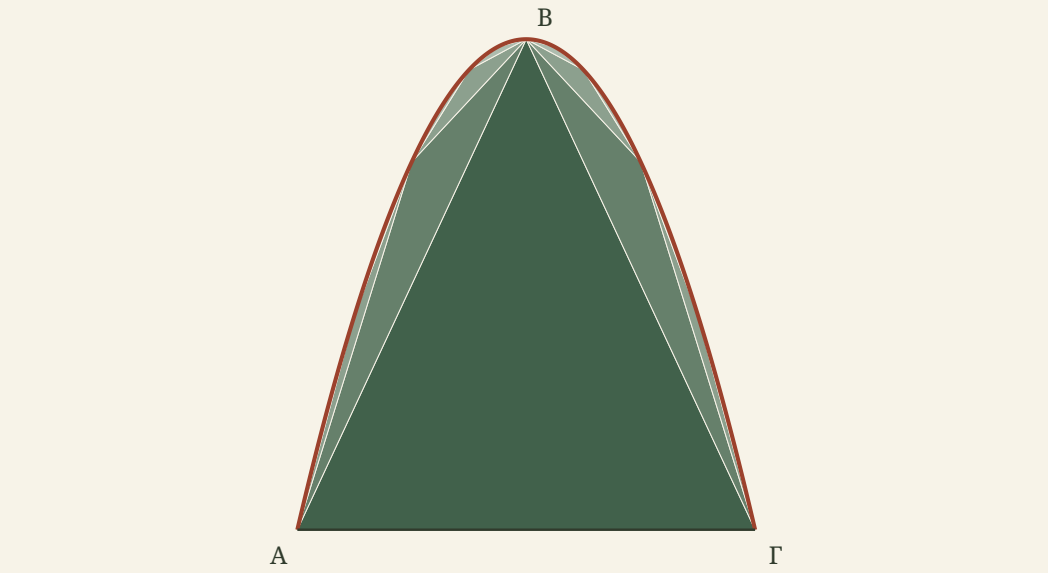
A third more.

Balancing sections suggests an area. Finite triangles and shrinking bounds supply a different kind of argument.

Your experiment



Geometry

Chord 7/3; height 5/2; first triangle area 35/12; large auxiliary triangle area 35/3.

Selected cut

u = 1/2 from A to Γ. Parabola section 5/2; triangle section 5. These are lengths.

Trial moments

Left 35/6; right 35/6; difference 0. Common section thickness and area density suppressed. The separate beam uses a left arm 1 times KΓ.

What this cut establishes

The selected interior cut balances. The arm identity holds for every interior cut.

The finite account

Complete layers; exact areas in square units

Layer	Triangles	Each area	Whole layer
0	1	35/12	35/12
1	2	35/96	35/48
2	4	35/768	35/192
3	8	35/6144	35/768
4	16	35/49152	35/3072

Inscribed sum
11935/3072

One third of the last layer
35/9216

Sum plus that third
35/9 = 4/3 × 35/12

Tangent upper bound
23905/6144

Gap between bounds
35/6144 = 1/512 of the first triangle

Discovery and proof remain different

The curved segment is drawn as y = 4hu(1-u) across a chord w, with x = w(u - 1/2). The auxiliary tangent triangle has four times the first inscribed triangle's area. At an interior cut, the transferred section and the triangle section balance exactly when the left arm equals КГ. Both endpoint moments vanish and cannot identify that arm. Our separate horizontal beam preserves the source's projected arm ratios; it is a new instructional diagram.

The mechanical comparison suggests 4/3 of the first triangle's area, using the large triangle's centroid. The source says the preceding reasoning has not proved its conclusion. A finite set of balanced slices does not itself prove the area result.

In the fill view, each child triangle has one-eighth of its parent's area, so a complete next layer has one-quarter of the previous layer. Adding one-third of the final complete layer to the finite sum gives 4/3 of the first triangle. That correction is not another region filled in the drawing.

Our modern tangent covers give strict lower and upper bounds. Their gap is A/(2 × 4^n), where A is the first triangle and n the final layer. Any proposed positive difference between the candidate and the true area can be made larger than this gap by enough finite refinements. Both values remain between the same bounds, a contradiction. This

coordinate exhaustion argument is distinct from the printed Quadrature XXIV, whose comparisons have their own source diagram.

Read the conditions

Greek and Latin follow the retained edited print. Line-end hyphens are joined, typographic figure letters normalized to Greek capitals, and inline footnote numbers omitted. Words and mathematical formulas are otherwise retained. New English and notes are teaching aids without independent philological review. The Method excerpt is not a full volume; the 1881 Quadrature volume predates discovery of the palimpsest.

Method I, printed pp. 434–435

A third greater than the triangle

Greek

λέγω, ὅτι ἐπίτριτόν ἐστι τὸ ΑΒΓ τμῆμα τοῦ ΑΒΓ τριγώνου.

Heiberg's Latin

Dico, segmentum ΑΒΓ tertia parte maius esse triangulo ΑΒΓ.

New English

I say that the segment ΑΒΓ is one-third greater than the triangle ΑΒΓ.

ἐπίτριτον gives the segment as 4/3 of the small inscribed triangle. The inverse comparison is 3/4: the triangle is three-quarters of the segment. Heiberg's tertia parte maius says 'greater by a third', not 'equal to one third'. Point letters identify the source figure; they are normalized to Greek capitals here.

Method I, printed pp. 436–437

One proportion sets the balance

Greek

ὡς ἄρα ἡ ΘΚ πρὸς ΚΝ, οὕτως ἡ ΜΞ πρὸς ΞΟ.

Heiberg's Latin

erit ΘΚ : ΚΝ = ΜΞ : ΞΟ.

New English

Thus, as ΘK is to KN, so MΞ is to ΞO.

ὡς … οὕτως pairs the two ratios: ‘as … so’. The longer triangle section MΞ stays at N; the shorter parabola section ΞO is transferred to Θ. Inverse weights and lever arms balance. Our separate horizontal beam preserves these arm ratios; it is a modern working diagram, not a manuscript facsimile.

Method I, printed pp. 438–439

The centre is a third along the arm

Greek

τριπλασία δέ ἐστιν ἡ ΘK τῆς KX· τριπλάσιον ἄρα καὶ τὸ AZΓ τρίγωνον τοῦ ABΓ τμήματος.

Heiberg’s Latin

verum ΘK = 3 KX; quare etiam AZΓ triangulus segmento ABΓ triplo maior est.

New English

But ΘK is three times KX; therefore the triangle AZΓ is also three times the segment ABΓ.

The centroid X of the large auxiliary triangle lies one-third of the way from K toward Γ. That large triangle is four times the small inscribed one. Combining ‘four times’ with ‘three times’ gives 4/3. The centroid theorem and mechanical comparison are inputs; a moving indicator does not prove them.

Printed opening of Method II, pp. 438–439; closes the preceding parabola argument

The author withholds the proof

Greek

Τοῦτο δὴ διὰ μὲν τῶν νῦν εἰρημένων οὐκ ἀποδέδεικται, ἔμφασιν δέ τινα πεποίηκε τὸ συμπέρασμα ἀληθὲς εἶναι·

Heiberg’s Latin

Hoc igitur per ea, quae nunc diximus, demonstratum illud quidem non est, at significationem quandam dedit, conclusionem veram esse;

New English

This has not been proved by what has just been said, but it has given some indication that the conclusion is true.

The printed II heading stands above this qualification. Its ‘this’ closes the preceding parabola argument before the sphere result begins. οὐκ negates ‘is demonstrated’; an exact balance should not silently remove that qualification. The perfect passive ἀποδέδεικται asks what has actually been demonstrated. The source’s qualification remains beside the mechanical result. The excerpt follows the retained print; it does not claim a new manuscript decipherment.

Quadrature XXI, Heiberg 1881, printed pp. 344–345

Each small triangle is an eighth

Greek

δεικτέον, ὅτι ὀκταπλάσιόν ἐστι τὸ ΑΒΓ τρίγωνον τοῦ ΑΒΖ τριγώνου.

Heiberg’s Latin

demonstrandum est, esse ΑΒΓ = 8 ΑΒΖ.

New English

It must be shown that the triangle ΑΒΓ is eight times the triangle ΑΒΖ.

The two newly inserted triangles each have one-eighth of their parent’s area. Together they have one-quarter. A count of triangles doubles while their combined area is quartered. The formula in the Latin retains its mathematical statement; point letters are normalized. The fill diagram groups both children as one complete layer.

Quadrature XXIII, Heiberg 1881, printed pp. 346–347

Add a third of the smallest

Greek

Εἴ κα μεγέθεα τεθέωντι ἐξῆς ἐν τῷ τετραπλασίονι λόγῳ, τὰ πάντα μεγέθεα καὶ ἔτι τοῦ ἐλαχίστου τὸ τρίτον μέρος ἐς τὸ αὐτὸ συντεθέντα ἐπίτριτα ἐσσοῦνται τοῦ μεγίστου.

Heiberg’s Latin

Si magnitudines quaedam ponuntur in quadrupla deinceps proportione, omnes magnitudines et praeterea tertia pars minimae simul sumptae tertia parte maiores erunt maxima.

New English

If magnitudes are set one after another in the fourfold ratio, all the magnitudes together, with one-third of the smallest added, will be one-third greater than the greatest.

This is a finite statement. ‘The smallest’ is the last complete layer in the account, not one individual triangle within that layer. The extra third is an algebraic correction; it has not filled the remaining curved pieces. εἴ κα is the conditional form in this edited Doric text; it is not silently changed to the Method’s wording.

Quadrature XXIV, Heiberg 1881, printed pp. 352–353

Neither smaller nor larger

Greek

οὐκ ἄρα τὸ ΑΔΒΕΓ τμᾶμα ἔλασσόν ἐστι τοῦ Κ χωρίου. ἐδείχθη δέ, ὅτι οὐδὲ μεῖζον. ἴσον ἄρα ἐστὶν τῷ Κ.

Heiberg’s Latin

itaque segmentum ΑΔΒΕΓ minus non est spatio K. demonstratum autem est, id ne maius quidem esse. itaque spatio K aequale est.

New English

Therefore the segment ΑΔΒΕΓ is not smaller than the area K. It has also been shown that it is not larger. It is therefore equal to K.

The printed argument rules out both strict inequalities. A finite drawing has not become the curve: the proof concerns what every sufficiently refined construction can establish. K here labels the comparison area, while K in the Method labels the fulcrum. The two source figures must not be silently treated as one. Our tangent outer covers are a separate coordinate demonstration of shrinking bounds, not a transcription of this proof.

Words doing mathematical work

ἐπίτριτον
epitriton — a third greater; in the ratio 4:3. The prefix ἐπι- adds a third to a whole. Latin tertia parte maius makes the comparison explicit.

τμῆμα · τμᾶμα
tmēma · tmāma — the cut-off segment. The Method and the edited Doric Quadrature show different forms. The word names the curved segment, not one of our vertical slices.

τριγώνου
trigōnou — of the triangle. A genitive after the ratio word: this triangle is the comparison quantity. Latin uses triangulo after maius.

ὡς … οὕτως
hōs … houtōs — as … so. The paired words match the arm ratio to the inverse section-length ratio. They are a proportion, not a time sequence.

ἰσορροπήσει
isorropēsei — will balance. The future verb in the surrounding passage describes equal turning moments. Equal weights are not required when arms differ.

κέντρον βάρους
kentron barous — centre of weight. βάρος is weight; βάρους is ‘of weight’. Latin centrum gravitatis supplies the corresponding genitive.

τριπλασία
triplasia — threefold. Here it compares the long arm with the centroid arm. The resulting large-triangle area is threefold too.

οὐκ ἀποδέδεικται
ouk apodedeiktai — is not demonstrated. The negation belongs to the claim about proof. The balance gives an indication while leaving a geometrical argument to supply.

ἔμφασιν
emphasin — an indication. In this sentence it names what the mechanical reasoning has supplied. It does not retract οὐκ, ‘not’.

ὀκταπλάσιον
oktaplasion — eightfold. One parent is eight times either child. Two children together therefore make one-quarter of the parent.

ἑξῆς
hexēs — one after another. The layer account is ordered: each complete area is followed by a quarter of itself. The word does not itself assert an infinite sum.

τοῦ ἐλαχίστου
tou elachistou — of the smallest. The article and superlative identify the final magnitude in the finite list. It is one whole layer here.

τὸ τρίτον μέρος
to triton meros — the third part. The last layer’s area divided by three, not a third of the first triangle. Latin tertia pars minimae carries both ideas.

ἔλασσον · μεῖζον

elasson · meizon — smaller · larger. The final proof excludes each alternative. Our inner and outer bounds show a related exhaustion structure in modern coordinates.

Complete source pages

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τοῦ ἄξονος διαιρεθέντος | ρῦτως, ὥστε τὸ πρὸς τῇ κορυφῇ τμῆ|μα τριπλάσιον εἶναι τοῦ λοιποῦ.

Χρη|σόμεθα δὲ καὶ [ἐν τῷ προγεγραμ|μένῳ Κωνοει- 11
δῶν] τῷδε τῷ θεωρή|ματι· Ἐὰν ὁποσαοῦν μεγέθη
5 ἄλ|λοις μεγέθεσιν ἴσοις τὸ πλῆθος | κατὰ δύο τὸν αὐτὸν ἔχῃ λόγον τὰ ὁ|μοίως τεταγμένα, ᾗ δὲ τὰ πρῶτα |
57ᵛ col. 1 μεγέθη πρὸς ἄλλα μεγέθη ἐν λόγοις ὁποιοισοῦν, ἢ τὰ | πάντα ἢ τινα αὐτῶν, καὶ τὰ ὑστε|ρον μεγέθη πρὸς ἄλλα μεγέθη τὰ ὁμόλογα ἐν | τοῖς αὐτοῖς λό-
10 γοις ᾗ, πάντα τὰ | πρῶτα μεγέθη πρὸς πάντα τὰ | λεγόμενα τὸν αὐτὸν ἔχει λόγον, | ὃν ἔχει πάντα τὰ ὑστερον πρὸς | πάντα τὰ λεγόμενα.

α'.

Ἔστω | τμῆμα τὸ ΑΒΓ περιεχόμενον | ὑπὸ εὐθείας
15 τῆς ΑΓ καὶ ὀρθο|γωνίου κώνου τομῆς τῆς ΑΒΓ, | καὶ τετμήσθω δίχα ἡ ΑΓ τῷ Δ, | καὶ παρὰ τὴν διάμετρον ἤχθω ἡ | ΔΒΕ, καὶ ἐπεζεύχθωσαν αἱ ΑΒ, | ΒΓ.

λέγω, ὅτι ἐπίτριτόν ἐστιν τὸ ΑΒΓ | τμῆμα τοῦ ΑΒΓ τριγώνου.

20 ἤχθω|σαν ἀπὸ τῶν Α, Γ σημείων ἡ μὲν | ΑΖ παρὰ τὴν ΔΒΕ, ἡ δὲ ΓΖ ἐπιψαύ|ουσα τῆς τομῆς, καὶ ἐκ-
64ʳ col. 1 βεβλήσ|⟨θω ἡ ΓΒ ἐπὶ τὸ Κ, καὶ κείσθω τῇ ΓΚ | ἴση ἡ ΚΘ⟩. νοείσθω ζυγὸς ὁ ΓΘ καὶ | μέσον αὐτοῦ τὸ Κ καὶ τῇ ΕΔ πα|ράλληλος τυχοῦσα ἡ ΜΞ.

α'] Hero, Metr. p. 80, 17 sqq. (ἐν τῷ Ἐφοδικῷ); 84, 11 sqq.

3 ἐν — Κωνοειδῶν] deleo. 4 τῷ] om. 5 ἴσοις] ἴσα. 7 πρὸς ἄλλα μεγέθη] om. λόγοις] τόποις. 9 ἄλλα μεγέθη] om. 13 α'] om.

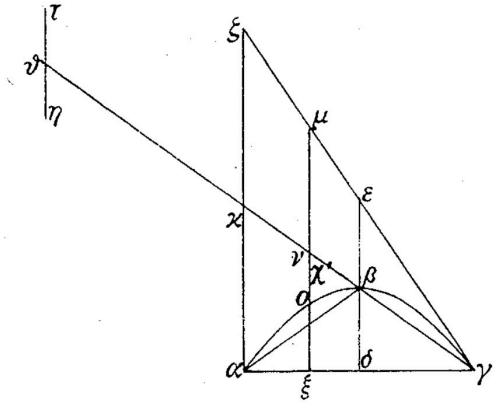
1) Demonstratur De conoid. et sphaeroid. 1. sed uerba lin. 3 ἐν τῷ προγεγραμμένῳ Κωνοειδῶν et forma neglegentiore et collocatione praua interpolata esse monstrantur; e margine irrepserunt.

11. Utemur autem hac quoque propositione:[1])

si quotlibet magnitudines ad alias magnitudines numero aequales binae ad binas, quae eodem loco positae sunt, eandem rationem habent, et priores magnitudines ad alias magnitudines quaslibet rationes habent aut omnes aut aliquot earum, posteriores autem magnitudines ad alias magnitudines, quae eodem loco positae sunt, easdem rationes habent, omnes magnitudines priores ad omnes, quae cum iis in proportione sunt, eandem rationem habent, quam omnes posteriores ad omnes, quae cum iis in proportione sunt.

I.

Sit segmentum $AB\Gamma$ comprehensum recta $A\Gamma$ sectioneque coni rectanguli $AB\Gamma$, et $A\Gamma$ in Δ in duas partes aequales



secetur, diametro autem parallela ducatur ΔBE, et ducantur AB, $B\Gamma$.

Dico, segmentum $AB\Gamma$ tertia parte maius esse triangulo $AB\Gamma$.

A punctis A, Γ ducantur AZ rectae ΔBE parallela, ΓZ autem sectionem contingens, et ΓB ad K producatur, ponaturque $K\Theta = \Gamma K$. recta $\Gamma\Theta$ libra fingatur punctumque eius medium K, et $M\Xi$ recta aliqua rectae $E\Delta$ parallela.

Method I and its closing warning, printed pp. 434–435, Greek and Heiberg's Latin. Whole scanned opening. PDF page 5. SHA-256 dbe749a6caf3f317469af78c14eafae8c172e448bc8691cad887349785eb0f84.

ἐπεὶ οὖν | παραβολή ἐστιν ἡ ΓΒΔ, καὶ ἐφά|πτεται
ἡ ΓΖ, καὶ τεταγμένως ἡ | ΓΔ, ἴση ἐστὶν ἡ ΕΒ τῇ
ΒΔ· τοῦτο γὰρ ἐν | τοῖς στοιχείοις δείκνυται· διὰ δὴ |
τοῦτο, καὶ διότι παράλληλοί εἰσιν | αἱ ΖΑ, ΜΞ τῇ
ΕΔ, ἴση ἐστὶν καὶ ἡ | μὲν ΜΝ τῇ ΝΞ, ἡ δὲ ΖΚ τῇ
ΚΑ. | καὶ ἐπεί ἐστιν, ὡς ἡ ΓΑ πρὸς ΑΞ, οὕ|τως ἡ
ΜΞ πρὸς ΞΟ [τοῦτο γὰρ ἐν | λήμματι δείκνυται], ὡς
δὲ ἡ ΓΑ πρὸς | ΑΞ, οὕτως ἡ ΓΚ πρὸς ΚΝ, καὶ
ἴση | ἐστὶν ἡ ΓΚ τῇ ΚΘ, ὡς ἄρα ἡ ΘΚ | πρὸς ΚΝ,
57v col. 2 οὕτως ἡ ΜΞ πρὸς ΞΟ. | καὶ ἐπεὶ τὸ Ν σημεῖον κέντρον |
11 τοῦ βάρους τῆς ΜΞ εὐθείας ἐστίν, | ἐπεί ἴση ἐστὶν
ἡ ΜΝ τῇ ΝΞ, | ἐὰν ἄρα τῇ ΞΟ ἴσην θῶμεν τὴν
ΤΗ | καὶ κέντρον τοῦ βάρους αὐτῆς τὸ | Θ, ὅπως ἴση
ᾖ ἡ ΤΘ τῇ ΘΗ, ἰσορρο|ροπήσει ἡ ΤΘΗ τῇ ΜΞ αὐ-
15 τοῦ με|νούσῃ διὰ τὸ ἀντιπεπονθότως | τετμῆσθαι τὴν
ΘΝ τοῖς ΤΗ, ΜΞ | βάρεσιν, καὶ ὡς τὴν ΘΚ πρὸς
ΚΝ, | οὕτως τὴν ΜΞ πρὸς τὴν ΗΤ· ὥσ|τε τοῦ ἐξ ἀμφο-
τέρων βάρους κέν|τρον ἐστὶν τοῦ βάρους τὸ Κ. δμοί-|
ως δὲ καί, ὅσαι ἂν ἀχθῶσιν | ἐν τῷ ΖΑΓ τριγώνῳ
20 παράλλη|λοι τῇ ΕΔ, ἰσορροπήσουσιν ἀὐ|τοῦ μένουσαι
ταῖς ἀπολαμβα|νομέναις ἀπ' αὐτῶν ὑπὸ τῆς | τομῆς
μετενεχθείσαις ἐπὶ τὸ | <Θ, ὥστε εἶναι τοῦ ἐξ ἀμφο-
64r col. 2 τέ>|ρων κέντρον τοῦ βάρους τὸ Κ. | καὶ ἐπεὶ ἐκ μὲν
τῶν ἐν τῷ ΓΖΑ | τριγώνῳ τὸ ΓΖΑ τρίγωνον συν-
25 έστηκεν, ἐκ δὲ τῶν | ἐν τῇ τομῇ ὁμοίως τῇ ΞΟ λαμ-|
βανομένων συνέστηκε τὸ ΑΒΓ | τμῆμα, ἰσορροπήσει
ἄρα τὸ | ΖΑΓ τρίγωνον αὐτοῦ μένον τῷ | τμήματι τῆς
τομῆς τεθέν|τι περὶ κέντρον τοῦ βάρους τὸ Θ | κατὰ
τὸ Κ σημεῖον, ὥστε τοῦ ἐ|ξ ἀμφοτέρων κέντρον εἶ-
30 ναι | τοῦ βάρους τὸ Κ. τετμήσθω δὴ | ἡ ΓΚ τῷ Χ,
ὥστε τριπλασίαν | εἶναι τὴν ΓΚ τῆς ΚΧ· ἔσται ἄρα |

iam quoniam $\Gamma B\Delta$ parabola[1]) est, ΓZ autem contingit et ordinate ducta est $\Gamma\Delta$, erit $EB = B\Delta$; hoc enim in elementis[2]) demonstratur; ea igitur de causa, et quia ZA, $M\Xi$ rectae $E\Delta$ parallelae sunt, erit etiam [Eucl. VI, 4; V, 9] $MN = N\Xi$ et $ZK = K\Delta$. et quoniam est

$$\Gamma A : A\Xi = M\Xi : \Xi O,^{3})\ \text{et}\ \Gamma A : A\Xi = \Gamma K : KN$$

[Eucl. VI, 2; V, 18], et $\Gamma K = K\Theta$, erit $\Theta K : KN = M\Xi : \Xi O$. et quoniam punctum N centrum grauitatis est rectae $M\Xi$ [lemm. 4], quia $MN = N\Xi$, si ponimus $TH = \Xi O$ et Θ centrum grauitatis eius, ita ut sit $T\Theta = \Theta H$ [lemm. 4], recta $T\Theta H$ cum recta $M\Xi$ suo loco manenti aequilibritatem seruabit, quia ΘN in contraria ratione ponderum TH, $M\Xi$ secta est, et $\Theta K : KN = M\Xi : HT$ [De plan. aequil. I, 6—7]; quare magnitudinis ex utraque compositae centrum grauitatis est K [lemm. 3]. similiter autem, quotcunque in triangulo $ZA\Gamma$ rectae $E\Delta$ parallelae rectae ducuntur, ipsae quoque suo loco manentes aequilibritatem seruabunt cum partibus suis, quae sectione abscinduntur, ad Θ ita transpositis, ut magnitudinis ex utraque compositae centrum grauitatis sit K. et quoniam ex rectis in triangulo ΓZA ductis triangulus ΓZA constat, ex rectis autem in sectione eodem modo, quo ΞO, sumptis segmentum $AB\Gamma$, triangulus $ZA\Gamma$ suo loco manens cum segmento sectionis circum Θ centrum grauitatis posito aequilibritatem seruabit in puncto K, ita ut magnitudinis ex utraque compositae centrum grauitatis sit K. puncto igitur X recta ΓK ita diuidatur, ut sit $\Gamma K = 3KX$; itaque X centrum grauitatis erit trianguli

1) Hoc quoque uestigium est interpolationis; Archimedes scripserat ὀρθογωνίου κώνου τομά.

2) Sc. conicorum Euclidis et Aristaei; est Quadrat. parab. 2.

3) Est Quadrat. parab. 5 coll. Eucl. V, 18. uerba τοῦτο — δείκνυται lin. 7 interpolatoris sunt; lemma, quod idem sine dubio in mg. addiderat, nunc non comparet.

6 ΓΑ] ΓΔ. 7 ΞΟ] ΞΘ. τοῦτο — δείκνυται] deleo. 14 ᾖ] om. 19 ἂν] ἐὰν. 20 ΕΔ] ΗΔ. μένουσαι] μενούσαις. 23 κέντρον] κέντρων. 24 τὸ ΓΖΑ τρίγωνον] om.

Method I and its closing warning, printed pp. 436–437, Greek and Heiberg's Latin. Whole scanned opening. PDF page 6. SHA-256 ab30628550231b3887a12ca8a9be7ec85a6ee8e4d5ea1f25352b6f7c8bc10f49.

τὸ X σημεῖον κέντρον βάρους | τοῦ $AZ\Gamma$ τριγώνου·
66r col. 1 δέδεικται γὰρ | ἐν τοῖς Ἰσορροπικοῖς. ἐπεὶ οὖν ἰ|σόρροπον τὸ $Z\Lambda\Gamma$ τρίγωνον αὐ|τοῦ μένον τῷ $B\Lambda\Gamma$ τμήματι κατὰ | τὸ K τεθέντι περὶ τὸ Θ κέντρον | τοῦ βάρους, καί ἐστιν τοῦ $Z\Lambda\Gamma$ τρι|γώνου κέντρον βάρους τὸ X, ἔστιν | ἄρα, ὡς τὸ $AZ\Gamma$ τρίγωνον πρὸς | τὸ $AB\Gamma$ τμῆμα κείμενον περὶ τὸ | Θ κέντρον, οὕτως ἡ ΘK πρὸς XK. | τριπλασία δέ ἐστιν ἡ ΘK τῆς KX· τρι|πλάσιον ἄρα καὶ τὸ $AZ\Gamma$ τρίγωνον | τοῦ $AB\Gamma$ τμήματος. ἔστι δὲ καὶ | τὸ $Z\Lambda\Gamma$ τρίγωνον τετραπλάσιον | τοῦ $AB\Gamma$ τριγώνου διὰ τὸ ἴσην εἶναι | τὴν μὲν ZK τῇ $K\Lambda$, τὴν δὲ $\Lambda\Delta$ τῇ | $\Delta\Gamma$· ἐπίτριτον ἄρα ἐστὶν τὸ $AB\Gamma$ τμῆ|μα τοῦ $AB\Gamma$ τριγώνου. [τοῦτο οὖν | φανερόν ἐστιν].

71v col. 1 β'.

Τοῦτο δὴ διὰ μὲν τῶν νῦν εἰρημένων | οὐκ ἀποδέδεικται, ἔμφασιν δέ | τινα πεποίηκε τὸ συμπέρασμα | ἀληθὲς εἶναι· διόπερ ἡμεῖς ὁ|ρῶντες μὲν οὐκ ἀποδεδειγμέ|νον, ὑπονοοῦντες δὲ τὸ συμπέ|ρασμα ἀληθὲς εἶναι, τάξο|μεν τὴν γεωμετρουμένην ἀ|πόδειξιν ἐξευρόντες αὐτοὶ τὴν | ἐκδοθεῖσαν πρότερον.

66r col. 2 Ὅτι δὲ πᾶ|σα σφαῖρα τετραπλασία ἐστὶν τοῦ | κώνου τοῦ βάσιν μὲν ἔχοντος | ἴσην τῷ μεγίστῳ κύκλῳ τῶν ἐν | τῇ σφαίρᾳ, ὕψος δὲ ἴσον τῇ ἐκ τοῦ | κέντρου τῆς σφαίρας, καὶ ὁ κύλιν|δρος ὁ βάσιν μὲν ἔχων ἴσην τῷ | μεγίστῳ κύκλῳ τῶν ἐν τῇ σφαίρᾳ, | ὕψος δὲ ἴσον τῇ διαμέτρῳ τῆς σφαί|ρας, ἡμιόλιος τῆς σφαίρας ἐστίν, | ὧδε θεωρεῖται κατὰ τρόπον τόνδε·

Ἔστω γάρ τις σφαῖρα, ἐν ᾗ μέγιστος | κύκλος ὁ $AB\Gamma\Delta$, διάμετροι δὲ αἱ | $A\Gamma$, $B\Delta$ πρὸς ὀρθὰς ἀλλή-

$AZ\Gamma$; hoc enim in libro De aequilibriis[1]) demonstratum est. quoniam igitur triangulus $Z\Lambda\Gamma$ suo loco manens in puncto K cum segmento $B\Lambda\Gamma$ circum Θ centrum grauitatis posito aequilibritatem seruat, et X centrum grauitatis est trianguli $Z\Lambda\Gamma$, erit, ut triangulus $AZ\Gamma$ ad segmentum $AB\Gamma$ circum Θ centrum positum, ita $\Theta K : XK$. uerum $\Theta K = 3\,KX$; quare etiam $AZ\Gamma$ triangulus segmento $AB\Gamma$ triplo maior est. sed triangulus $Z\Lambda\Gamma$ idem triangulo $AB\Gamma$ quadruplo maior est [ZMP. XXIV p. 179 nr. 7], quia $ZK = K\Lambda$ et $\Lambda\Delta = \Delta\Gamma$;[2]) ergo segmentum $AB\Gamma$ triangulo $AB\Gamma$ tertia parte maius est.[3])

II.

Hoc igitur per ea, quae nunc diximus, demonstratum illud quidem non est, at significationem quandam dedit, conclusionem ueram esse; quare, cum intellegamus, conclusionem demonstratam non esse, suspicemur autem, eam ueram esse, demonstrationem per geometriam a nobis ipsis inuentam suo loco[4]) proponemus, quam eandem antea edidimus.

Omnem uero sphaeram quadruplo maiorem esse cono basim habenti circulo maximo sphaerae aequalem, altitudinem autem radio sphaerae aequalem, et cylindrum basim habentem circulo maximo sphaerae aequalem, altitudinem autem diametro sphaerae aequalem, dimidia parte maiorem esse sphaera, per hanc methodum sic examinatur:

Sit enim sphaera, cuius circulus maximus $AB\Gamma\Delta$, dia-

1) Cfr. lemm. 5, et u. De plan. aequil. I, 15; II, 5. ZMP. XXIV p. 179 nr. 6a.

2) Nam $\Delta\Gamma : A\Gamma = \Delta B : AK$ (Eucl. VI, 4), unde
$$\Delta B = \tfrac{1}{2} AK = \tfrac{1}{4} AZ.$$

3) τοῦτο — ἐστιν lin. 13—14 cum lin. 16 sqq. stare nequeunt.

4) Sine dubio in extremo libro.

2 ἐπεὶ] ἔσται. 13 τοῦτο — 14 ἐστιν] deleo. 15 β'] om. Hic fig. p. 435. 20 τάξομεν] quod addubitat *Theodorus Reinach*, nunc praestare possum. 22 τετραπλασία] διπλασία.

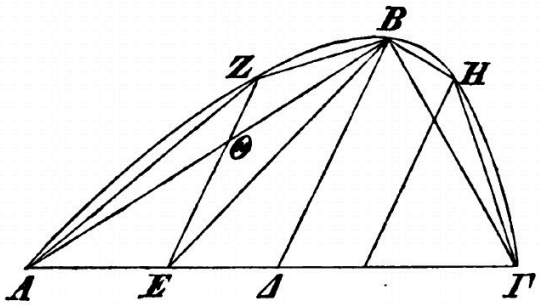
Method I and its closing warning, printed pp. 438–439, Greek and Heiberg's Latin. Whole scanned opening. PDF page 7. SHA-256 da3bc97c287b9c6ed39f82786dd624abf643f1e90da18790cb3c1c160509140f.

ἡμίσεος διὰ τοῦτο, φανερόν, ὅτι ἐλασσούντες ἀεὶ τὰ λειπόμενα τμάματα ποιήσομες ταῦτα ἐλάσσονα παντὸς τοῦ προτεθέντος χωρίου.

κα΄.

Εἴ κα εἰς τμᾶμα περιεχόμενον ὑπὸ εὐθείας καὶ ὀρθογωνίου κώνου τομᾶς τρίγωνον ἐγγραφῇ τὰν αὐτὰν βάσιν ἔχον τῷ τμάματι καὶ ὕψος τὸ αὐτό, ἐγγραφέωντι δὲ καὶ ἄλλα τρίγωνα ἐς τὰ λειπόμενα τμάματα τὰν αὐτὰν βάσιν ἔχοντα τοῖς τμαμάτεσσιν καὶ ὕψος τὸ αὐτό, ἑκατέρου τῶν τριγώνων τῶν εἰς τὰ περιλειπόμενα τμάματα ἐγγραφέντων ὀκταπλάσιον ἐσσείται τὸ τρίγωνον τὸ εἰς τὸ ὅλον τμᾶμα ἐγγραφέν.

ἔστω τὸ $AB\Gamma$ τμᾶμα, οἷον εἰρήται, καὶ τετμάσθω ἁ $A\Gamma$ δίχα τῷ Δ, ἁ δὲ $B\Delta$ ἄχθω παρὰ τὰν διάμετρον. τὸ B ἄρα σαμεῖον κορυφά ἐστιν τοῦ τμάματος. τὸ ἄρα $AB\Gamma$ τρίγωνον τὰν αὐτὰν βάσιν ἔχει τῷ τμάματι καὶ ὕψος τὸ αὐτό. πάλιν τετμάσθω δίχα ἁ $A\Delta$



τῷ E, καὶ ἄχθω ἁ EZ παρὰ τὰν διάμετρον, τετμάσθω δὲ ἁ AB κατὰ τὸ Θ. τὸ ἄρα Z σαμεῖον κορυφά ἐστι τοῦ τμάματος τοῦ AZB. τὸ δὴ AZB τρίγωνον τὰν

1. ημισους F, uulgo. 2. τμηματα F; corr. Torellius, ut lin. 5, 7, 8, 11, 12, 13, 15, 16. ποιησομεν F, uulgo. 9. τμη-

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 342. Complete printed page. PDF page 385. SHA-256 985e37725bd9b3dc30e4696d396101b08ded10fa002c2e6f1a10f163b0e6e7c0.

parte dimidia, abstulerimus, manifestum est, nos spatia reliqua semper minuentes [aliquando] ea minora facturos esse quouis spatio dato [Eucl. X, 1].

XXI.

Si segmento linea recta et coni rectanguli sectione comprehenso triangulus inscribitur eandem basim habens, quam segmentum, et altitudinem eandem, et etiam segmentis reliquis alii trianguli inscribuntur eandem basim habentes, quam segmenta, et altitudinem eandem, triangulus toti segmento inscriptus aequalis erit utriuis triangulorum segmentis reliquis inscriptorum octies sumpto.

sit $AB\Gamma$ segmentum, quale diximus, et linea $A\Gamma$ in puncto Δ in duas partes aequales diuidatur, et $B\Delta$ diametro parallela ducatur. itaque B punctum uertex est segmenti [prop. 18]. itaque triangulus $AB\Gamma$ eandem basim habet, quam segmentum, et altitudinem eandem.[1]) rursus linea $A\Delta$ in puncto E in duas partes aequales diuidatur, et diametro parallela ducatur EZ, et ab ea linea AB in Θ secetur. itaque punctum Z uertex est segmenti AZB.[2]) quare triangulus AZB eandem ba-

1) Nam altitudo trianguli linea est ab B ad $A\Gamma$ perpendicularis, quae eadem altitudo est segmenti, quia B uertex est (p. 336, 15).

2) Nam $E\Theta \neq B\Delta$; itaque $AE : E\Delta = A\Theta : \Theta B$; sed $AE = E\Delta$; quare $A\Theta = \Theta B$; tum u. prop. 18.

μασιν F; *τμηματεσσι* uulgo; *τμαματεσσι* Torellius. 11. *εσται* per comp. F, uulgo. 13. *τετμησθω* F; corr. Torellius, ut lin. 17, 18. 19. *σημειον* F. *κορυφη* F; corr. Torellius. 20. *τοῦ*] alterum suprascr. manu 1 F. *τμηματος* F.

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 343. Complete printed page. PDF page 386. SHA-256 08552523e097033d7f2640f9eef49c5697c37f95e7c7d5fa863a408792ae475a.

αὐτὰν βάσιν ἔχει τῷ *AZB* τμάματι καὶ ὕψος τὸ αὐτό. δεικτέον, ὅτι ὀκταπλάσιόν ἐστι τὸ *ABΓ* τρίγωνον τοῦ *ABZ* τριγώνου.

ἔστιν οὖν ἁ *BΔ* τᾶς μὲν *EZ* ἐπίτριτος, τᾶς δὲ *EΘ* διπλασία. διπλασία ἄρα ἐστὶν ἁ *EΘ* τᾶς *ΘZ*. ὥστε καὶ τὸ *AEB* τρίγωνον διπλάσιόν ἐστι τοῦ *ZBA*· τὸ μὲν γὰρ *AEΘ* διπλάσιόν ἐστι τοῦ *AΘZ*, τὸ δὲ *ΘBE* τοῦ *ZΘB*. ὥστε τὸ *ABΓ* τοῦ *AZB* ἐστι ὀκταπλάσιον. ὁμοίως δὲ δειχθησέται καὶ τοῦ εἰς τὸ *BHΓ* τμᾶμα ἐγγραφέντος.

κβ′.

Εἴ κα ᾖ τμᾶμα περιεχόμενον ὑπὸ εὐθείας καὶ ὀρθογωνίου κώνου τομᾶς, καὶ χωρία τεθέωντι ἐξῆς ὁποσαοῦν ἐν τῷ τετραπλασίονι λόγῳ, ᾖ δὲ τὸ μέγιστον τῶν χωρίων ἴσον τῷ τριγώνῳ τῷ βάσιν ἔχοντι τὰν αὐτὰν τῷ τμάματι καὶ ὕψος τὸ αὐτό, σύμπαντα τὰ χωρία ἐλάσσονα ἐσσείται τοῦ τμάματος.

ἔστω γὰρ τμᾶμα τὸ *AΔBEΓ* περιεχόμενον ὑπό τε εὐθείας καὶ ὀρθογωνίου κώνου τομᾶς, χωρία δὲ ἔστω ὁποσαοῦν ἐξῆς κείμενα τὰ *Z*, *H*, *Θ*, *I*, τετραπλάσιον δὲ ἔστω τὸ ἀγούμενον τοῦ ἑπομένου, μέγιστον δὲ ἔστω τὸ *Z*, καὶ ἔστω τὸ *Z* ἴσον τῷ τριγώνῳ τῷ βάσιν ἔχοντι τὰν αὐτὰν τῷ τμάματι καὶ ὕψος ἴσον. λέγω, ὅτι τὸ τμᾶμα τῶν *Z*, *H*, *Θ*, *I* χωρίων μεῖζόν ἐστιν.

ἔστω τοῦ μὲν ὅλου τμάματος κορυφὰ τὸ *B*, τῶν δὲ περιλειπομένων τμαμάτων τὰ *Δ*, *E*. ἐπεὶ οὖν τὸ *ABΓ* τρίγωνον ὀκταπλάσιόν ἐστιν ἑκατέρου τῶν *ABΔ*, *BEΓ*

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1. τῷ] το F. τμηματι F; corr. Torellius, ut lin. 12, 16, 17, 18, 23, 24, 25, 26. 10. τμηματος F; corr. B. 13. τομης

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 344. Complete printed page. PDF page 387. SHA-256 74472dd7ffddb018b6e4d1f98b8cb4866041cc1232a17c94d09fa9123be42846.

sim habet, quam segmentum AZB, et altitudinem eandem. demonstrandum est, esse $AB\Gamma = 8\,ABZ$.

iam est $B\Delta = \frac{4}{3}EZ$ [prop. 19], sed etiam

$$B\Delta = 2\,E\Theta.$$ [1])

itaque $E\Theta = 2\Theta Z$.[2]) quare etiam $AEB = 2ZBA$; nam $AE\Theta = 2A\Theta Z$ et $\Theta BE = 2Z\Theta B$ [Eucl. VI, 1]. quare est $AB\Gamma = 8AZB$.[3]) et eodem modo demonstrabimus, esse etiam $AB\Gamma = 8BH\Gamma$.

XXII.

Si datum est segmentum linea recta et coni rectanguli sectione comprehensum, et ponuntur spatia quotlibet, quae deinceps in quadrupla proportione sunt, et maximum spatium aequale est triangulo basim habenti eandem, quam segmentum, et altitudinem eandem, omnia simul spatia minora erunt segmento.

sit enim $A\Delta BE\Gamma$ segmentum recta linea et coni rectanguli sectione comprehensum, et ponantur quotlibet spatia deinceps, Z, H, Θ, I, et praecedens quadruplo maius sit sequenti, et maximum sit Z, et Z aequale sit triangulo basim eandem habenti, quam segmentum, et altitudinem aequalem. dico, segmentum maius esse spatiis Z, H, Θ, I.

totius segmenti uertex sit B, et segmentorum reliquorum uertices Δ, E. quoniam igitur

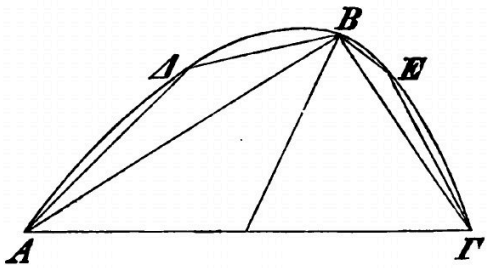
$$AB\Gamma = 8\,AB\Delta = 8\,BE\Gamma$$ [prop. 21],

1) Nam $AE : A\Delta = E\Theta : B\Delta = 1 : 2$.
2) Nam $E\Theta = \frac{2}{3}EZ$; itaque $\Theta Z = \frac{1}{3}EZ$.
3) Nam $AB\Gamma = 2\,AB\Delta = 2(ABE + BE\Delta) = 4\,ABE$.

F. 17. *εσται* per comp. F, uulgo. 19. *κώνου*] om. F, corr. V. 21. *ηγουμενον* F, uulgo. 26. *ἐπεί*] επι F.

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 345. Complete printed page. PDF page 388. SHA-256 abed7db1c016465afe72c5604f41277231c4fb93a8f619a66ceb97a6cbb3e8cd.

τριγώνων, δῆλον, ὅτι [ὡς] ἀμφοτέρων αὐτῶν ἐστι τετραπλάσιον. καὶ ἐπεὶ τὸ $AB\Gamma$ τρίγωνον ἴσον ἐστὶ τῷ Z χωρίῳ, κατὰ τὰ αὐτὰ δὲ καὶ τὰ $A\Delta B$, $BE\Gamma$ τρίγωνα ἴσα ἐστὶ τῷ H χωρίῳ, ὁμοίως δὲ δειχθησέται



καὶ τὰ εἰς τὰ περιλειπόμενα τμάματα ἐγγραφόμενα τρίγωνα τὰν αὐτὰν βάσιν ἔχοντα τοῖς τμαμάτεσσιν καὶ ὕψος τὸ αὐτὸ ἴσα ἐόντα τῷ Θ, καὶ τὰ ἐς τὰ ὕστερον γενόμενα τμάματα ἐγγραφόμενα τρίγωνα ἴσα τῷ I χωρίῳ, σύμπαντα ἄρα τὰ προτεθέντα χωρία ἴσα ἐσσούνται πολυγώνῳ τινὶ ἐγγραφέντι εἰς τὸ τμᾶμα. φανερὸν οὖν, ὅτι ἐλάσσονά ἐστι τοῦ τμάματος.

κγ'.

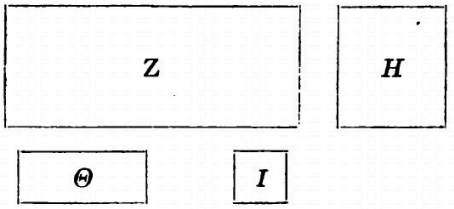
Εἴ κα μεγέθεα τεθέωντι ἑξῆς ἐν τῷ τετραπλασίονι λόγῳ, τὰ πάντα μεγέθεα καὶ ἕτι τοῦ ἐλαχίστου τὸ τρίτον μέρος ἐς τὸ αὐτὸ συντεθέντα ἐπίτριτα ἐσσούνται τοῦ μεγίστου.

ἔστω οὖν ὁποσαοῦν μεγέθεα ἑξῆς κείμενα τὰ A, B, Γ, Δ, E τετραπλασίονα ἕκαστον τοῦ ἑπομένου, μέγιστον δὲ ἔστω τὸ A. ἔστω δὲ τὸ μὲν Z τρίτον τοῦ B, τὸ δὲ H τοῦ Γ, το δὲ Θ τοῦ Δ, τὸ δὲ I τοῦ E.

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1. ὡς deleo. 3. τῷ] το F. 5. οτι και F, uulgo; ὅτι *deleui.* τμηματα F; corr. Torellius, ut lin. 8, 10, 11. 6. *τμημασιν F*, uulgo. 7. ἴσα ἐόντα τῷ] scripsi; ισων ωντων το

adparet, esse $AB\Gamma = 4(AB\Delta + BE\Gamma)$. et quoniam $AB\Gamma = Z$, et eodem modo $A\Delta B + BE\Gamma = H$, et[1]) similiter demonstrabimus, etiam triangulos reliquis segmentis inscriptos eandem basim habentes, quam



segmenta, et altitudinem eandem aequales esse spatio Θ, et triangulos segmentis deinde ortis inscriptos aequales spatio I, omnia igitur simul spatia data aequalia erunt polygono cuidam segmento inscripto. manifestum est igitur, minora ea esse segmento.

XXIII.

Si magnitudines quaedam ponuntur in quadrupla deinceps proportione, omnes magnitudines et praeterea tertia pars minimae simul sumptae tertia parte maiores erunt maxima.[2])

ponantur igitur deinceps quotlibet magnitudines A, B, Γ, Δ, E, singulae quadruplo maiores sequenti, et maxima sit A. sit autem $Z = \frac{1}{3}B$, $H = \frac{1}{3}\Gamma$, $\Theta = \frac{1}{3}\Delta$,

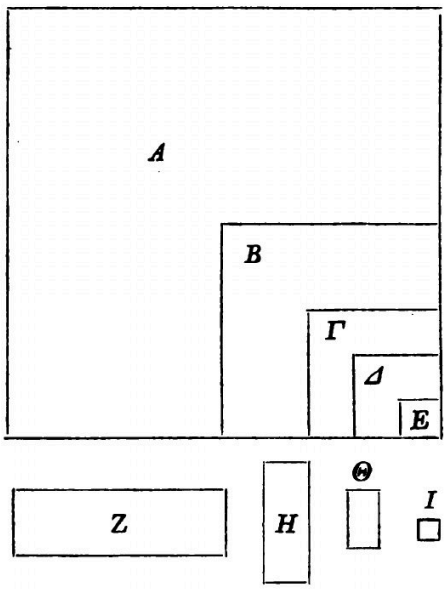
1) Lin. 3—4 suspicor, potius sic scribendum esse: *κατὰ ταῦτα δὴ καί . . . . χωρίῳ. ὁμοίως δή.*

2) Cfr. Quaest. Arch. p. 57—58. Figura aliter descripta est in F (u. infra).

(*ων* bis per comp.) F, uulgo; *ἴσα ἐντὶ τῷ* B, Torellius. *τὰ ἐς*] scripsi; *ες* F, uulgo. 9. *I*] $\overline{\text{Ϛωι}}$ F; corr. B. 11. *ἐλάσσονα*] scripsi; *ελασσον* F, uulgo. 13. *τεθέωντι*] scripsi; *συντεθεωντι* F; *συντιθέωντι* uulgo. *ἐξῆς ὁποσαοῦν* Torellius. 18. *τετραπλάσιον* Nizzius.

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 347. Complete printed page. PDF page 390. SHA-256 6ffd6721135c507254c55c2840b798e14790017465533f8468d8a9150b236751.

ἐπεὶ οὖν τὸ μὲν Z τοῦ B τρίτον μέρος ἐστίν, τὸ δὲ B τοῦ A τέταρτον μέρος ἐστίν, ἀμφότερα τὰ B, Z μέρος τρίτον ἐστὶ τοῦ A. διὰ τὰ αὐτὰ δὴ καὶ τὰ H, Γ τοῦ B, καὶ τὰ Θ, Δ τοῦ Γ, καὶ τὰ I, E τοῦ Δ. καὶ τὰ σύμπαντα δὴ τὰ B, Γ, Δ, E, Z, H, Θ, I τρίτον μέρος ἐστὶ τῶν συμπάντων τῶν A, B, Γ, Δ. ἐντὶ δὲ καὶ αὐτὰ τὰ Z, H, Θ τρίτον μέρος αὐτῶν τῶν B, Γ, Δ. καὶ τὰ λοιπὰ ἄρα τὰ B, Γ, Δ, E, I τοῦ λοιποῦ τρίτον μέρος ἐστὶ τοῦ A. δῆλον οὖν, ὅτι τὰ σύμπαντα τὰ A, B, Γ, Δ, E καὶ τὸ I, τουτέστι τὸ τρίτον τοῦ E, τοῦ A ἐστιν ἐπίτριτα.



κδ′.

Πᾶν τμᾶμα τὸ περιεχόμενον ὑπὸ εὐθείας καὶ ὀρθογωνίου κώνου τομᾶς ἐπίτριτόν ἐστι τριγώνου τοῦ τὰν αὐτὰν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον.

ἔστω γὰρ τὸ ΑΔΒΕΓ τμᾶμα περιεχόμενον ὑπὸ εὐθείας καὶ ὀρθογωνίου κώνου τομᾶς, τὸ δὲ ΑΒΓ τρίγωνον ἔστω τὰν αὐτὰν βάσιν ἔχον τῷ τμάματι καὶ

10. *H*] E F. *I*] Γ F. 11. *τῶν*] *του* per comp. F. 12. *Δ, E* F; corr. Torellius. 22. *τμημα* F; corr. Torellius, *ut lin. 25, 27.* 23. *την αυτην* F; corr. Torellius. 27. *αυτα* F.

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 348. Complete printed page. PDF page 391. SHA-256 538f9e5b30307caa353e3216fe650a6989e1b69a13f7c153b07571ec4bbbad6c.

$I = \frac{1}{3}E$. quoniam igitur est $Z = \frac{1}{3}B$ et $B = \frac{1}{4}A$, erit $B + Z = \frac{1}{3}A$. eadem de causa etiam erit

$$H + \Gamma = \frac{1}{3}B,\ \Theta + \Delta = \frac{1}{3}\Gamma,\ I + E = \frac{1}{3}\Delta.$$

erit igitur etiam

$$B + \Gamma + \Delta + E + Z + H + \Theta + I = \frac{1}{3}(A + B + \Gamma + \Delta).$$

est autem etiam $Z + H + \Theta = \frac{1}{3}(B + \Gamma + \Delta)$ [ex hypothesi]. quare etiam $B + \Gamma + \Delta + E + I = \frac{1}{3}A$. adparet igitur, esse $A + B + \Gamma + \Delta + E + I$, h. e.

$$A + B + \Gamma + \Delta + E + \frac{1}{3}E = \frac{4}{3}A.$$

XXIV.

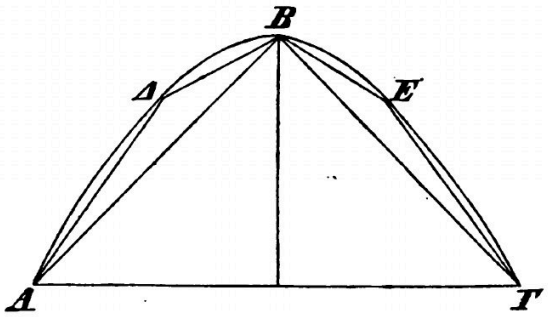
Quoduis segmentum linea recta et coni rectanguli sectione comprehensum tertia parte maius est triangulo eandem basim habenti, quam segmentum, et altitudinem aequalem.

sit enim $A\Delta BE\Gamma$ segmentum linea recta et coni rectanguli sectione comprehensum, et $AB\Gamma$ triangulus sit eandem basim habens, quam segmentum, et alti-

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 349. Complete printed page. PDF page 392. SHA-256 ea21264a3618a9cb15b4687b6696d2e01d17cb30be9d1f528b8547b93e9c2c43.

ὕψος ἴσον, τοῦ δὲ $AB\Gamma$ τριγώνου ἔστω ἐπίτριτον τὸ K χωρίον. δεικτέον, ὅτι ἴσον ἐστὶ τῷ $A\Delta BE\Gamma$ τμάματι.

εἰ γὰρ μή ἐστιν ἴσον, ἤτοι μεῖζόν ἐστιν ἢ ἔλασσον. ἔστω πρότερον, εἰ δυνατόν, μεῖζον τὸ $A\Delta BE\Gamma$ τμᾶμα



τοῦ K χωρίου. ἐνέγραψα δὴ τὰ $A\Delta B$, $BE\Gamma$ τρίγωνα, ὡς εἰρήται. ἐνεγραψα δὲ καὶ εἰς τὰ περιλειπόμενα τμάματα ἄλλα τρίγωνα τὰν αὐτὰν βάσιν ἔχοντα τοῖς τμαμάτεσσιν καὶ ὕψος τὸ αὐτό, καὶ ἀεὶ εἰς τὰ ὕστερον γινόμενα τμάματα ἐγγράφω [δύο] τρίγωνα τὰν αὐτὰν βάσιν ἔχοντα τοῖς τμαμάτεσσιν καὶ ὕψος τὸ αὐτό. ἐσσούνται δὴ τὰ καταλειπόμενα τμάματα ἐλάσσονα τᾶς ὑπεροχᾶς, ᾇ ὑπερέχει τὸ $A\Delta BE\Gamma$ τμᾶμα τοῦ K χωρίου. ὥστε τὸ ἐγγραφόμενον πολύγωνον μεῖζον ἐσσείται τοῦ K· ὅπερ ἀδύνατον. ἐπεὶ γάρ ἐστιν ἑξῆς κείμενα χωρία ἐν τῷ τετραπλασίονι λόγῳ, πρῶτον μὲν τὸ $AB\Gamma$ τρίγωνον τετραπλάσιον τῶν $A\Delta B$, $BE\Gamma$ τριγώνων, ἔπειτα δὲ αὐτὰ ταῦτα τετραπλάσια τῶν εἰς τὰ ἑπόμενα τμάματα ἐγγραφέντων, καὶ ἀεὶ οὕτω, δῆ-

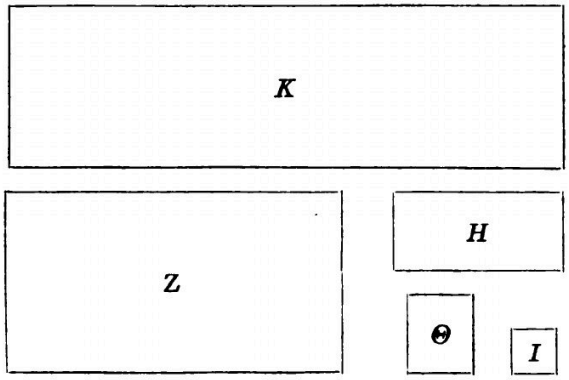
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2. τῷ] το F. 4. ἐστιν] (alterum) per comp. F. 5. $A\Delta EB\Gamma$ F; corr. Torellius, ut lin. 13. τμημα F; corr. Torellius, ut *lin. 3, 8, 10, 12, 13.* 6. $A\Delta B$] $A\Delta$ F. 9. τμημασιν F, uulgo, *ut lin. 11;* τμάμασι Torellius. 10. δύο] deleo. 12. εσουν-

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 350. Complete printed page. PDF page 393. SHA-256 cdc8493cd058412f1bee15a070db81067a40bfe85ced7b2db7572e030f45792d.

tudinem aequalem, et sit $K = \frac{4}{3}\Delta AB\Gamma$. demonstrandum est, esse $K = A\Delta BE\Gamma$.

nam si aequale non est, aut maius est aut minus. prius, si fieri potest, sit segmentum $A\Delta BE\Gamma$ maius



spatio K. inscripsi igitur triangulos $A\Delta B$, $BE\Gamma$ ita, ut diximus. et etiam segmentis reliquis alios triangulos inscripsi eandem basim habentes, quam segmenta, et altitudinem eandem, et semper segmentis deinde ortis triangulos inscribo eandem basim habentes, quam segmenta, et altitudinem eandem. erunt igitur [aliquando] segmenta reliqua minora excessu, quo segmentum $A\Delta BE\Gamma$ spatium K excedit [prop. 20 coroll.]. itaque polygonum inscriptum maius erit spatio K; quod fieri non potest. nam quoniam deinceps posita sunt spatia quaedam in quadrupla proportione, primum triangulus $AB\Gamma$ quadruplo maior triangulis $A\Delta B$, $BE\Gamma$ [prop. 21; cfr. p. 346, 1], deinde hi ipsi quadruplo maiores triangulis segmentis sequentibus inscriptis, et semper eodem modo, adparet, omnia

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ται F. 15. γάρ] addidi; om. F, uulgo. 18. αὐτὰ ταῦτα] scripsi; τα αυτα F, uulgo.

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 351. Complete printed page. PDF page 394. SHA-256 28a7cedc4126d8ca3cfca084c41e9042c6d3ff2345ba6ab595a39938f40dd53a.

λον, ὡς σύμπαντα τὰ χωρία ἐλάσσονά ἐστιν ἢ ἐπίτριτα τοῦ μεγίστου. τὸ δὲ K ἐπίτριτόν ἐστι τοῦ μεγίστου χωρίου. οὐκ ἄρα ἐστὶν μεῖζον τὸ ΑΔΒΕΓ τμᾶμα τοῦ K χωρίου. ἔστω δέ, εἰ δυνατόν, ἔλασσον. κείσθω δὴ τὸ μὲν ΑΒΓ τρίγωνον ἴσον τῷ Z, τοῦ δὲ Z τέταρτον τὸ H, καὶ ὁμοίως τοῦ H τὸ Θ, καὶ ἀεὶ ἑξῆς τιθέσθω, ἕως κα γενήται τὸ ἔσχατον ἔλασσον τᾶς ὑπεροχᾶς, ᾗ ὑπερέχει τὸ K χωρίον τοῦ τμάματος, καὶ ἔστω ἔλασσον τὸ I. ἔστιν δὴ τὰ Z, H, Θ, I χωρία καὶ τὸ τρίτον τοῦ I ἐπίτριτα τοῦ Z. ἔστιν δὲ καὶ τὸ K τοῦ Z ἐπίτριτον. ἴσον ἄρα τὸ K τοῖς Z, H, Θ, I καὶ τῷ τρίτῳ μέρει τοῦ I. ἐπεὶ οὖν τὸ K χωρίον τῶν μὲν Z, H, Θ, I χωρίων ὑπερέχει ἐλάσσονι τοῦ I, τοῦ δὲ τμάματος μείζονι τοῦ I, δῆλον, ὡς μείζονά ἐντι τὰ Z, H, Θ, I χωρία τοῦ τμάματος· ὅπερ ἀδύνατον. ἐδείχθη γάρ, ὅτι, ἐὰν ᾖ ὁποσαοῦν χωρία ἑξῆς κείμενα ἐν τετραπλασίονι λόγῳ, τὸ δὲ μέγιστον ἴσον ᾖ τῷ εἰς τὸ τμᾶμα ἐγγραφομένῳ τριγώνῳ, τὰ σύμπαντα χωρία ἐλάσσονα ἐσσείται τοῦ τμάματος. οὐκ ἄρα τὸ ΑΔΒΕΓ τμᾶμα ἔλασσόν ἐστι τοῦ K χωρίου. ἐδείχθη δέ, ὅτι οὐδὲ μεῖζον. ἴσον ἄρα ἐστὶν τῷ K. τὸ δὲ K χωρίον ἐπίτριτόν ἐστι τοῦ τριγώνου τοῦ ΑΒΓ. καὶ τὸ ΑΔΒΕΓ ἄρα τμᾶμα ἐπίτριτόν ἐστι τοῦ ΑΒΓ τριγώνου.

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3. τμημα F; corr. Torellius, ut lin. 8, 14, 15, 18, 19(?), 20, 23. 7. ἕως κα γενήται] scripsi; ωστε καταγενηται F, uulgo. 9. ἔλασσον] ἔσχατον Nizzius. δή] scripsi; δε F, uulgo. 11. τῷ] το F. 12. τῶν] τω F; corr. BC. 23. ἄρα] om. F; corr. Torellius. In fine F: Αρχιμηδ^{ε}ους τετραγωνισμος παραβολης· ευτυχοις λεον γεωμετρα: — + πολλοὺς ἐσ λυκάβαντας ἰοισ πολὺ φίλτατε μούσαις.

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 352. Complete printed page. PDF page 395. SHA-256 b9386d9e9eb2b56af47a172989a9aa201a6b3b5d1ed8dcded622abc0e4473862.

simul spatia minora esse quam tertia parte maiora maximo [prop. 23]. spatium autem K tertia parte maius est maximo spatio. itaque segmentum $A\Delta BE\Gamma$ maius non est spatio K. — sit autem, si fieri potest, minus. ponatur igitur $AB\Gamma = Z$[1]), $H = \frac{1}{4}Z$, $\Theta = \frac{1}{4}H$, et deinceps spatia ponantur, dum fiat ultimum spatium minus excessu, quo spatium K segmentum excedit [Eucl. X, 1], et [hoc excessu] minus sit I. sunt igitur

$$Z + H + \Theta + I + \tfrac{1}{3}I = \tfrac{4}{3}Z \text{ [prop. 23]};$$

sed erat etiam $K = \frac{4}{3}Z$. itaque

$$K = Z + H + \Theta + I + \tfrac{1}{3}I.$$

iam quoniam spatium K spatia Z, H, Θ, I excedit spatio minore, quam est spatium I, segmentum uero spatio maiore, quam est I, adparet, spatia Z, H, Θ, I maiora esse segmento; quod fieri non potest. nam demonstratum est, si spatia quotlibet deinceps data sint in quadrupla proportione, et maximum triangulo segmento inscripto aequale sit, omnia simul spatia minora fore segmento [prop. 22]. itaque segmentum $A\Delta BE\Gamma$ minus non est spatio K. demonstratum autem est, id ne maius quidem esse. itaque spatio K aequale est. sed spatium K tertia parte maius est triangulo $AB\Gamma$. itaque etiam segmentum $A\Delta BE\Gamma$ tertia parte maius est triangulo $AB\Gamma$.

1) Fortasse scribendum est lin. 4—5 *πείσθω δὴ τῷ μὲν ΑΒΓ τριγώνῳ ἴσον τὸ Ζ.*

Quadrature of the Parabola XXI–XXIV, Heiberg 1881, printed p. 353. Complete printed page. PDF page 396. SHA-256 493259826b4efc3318dee13b8dd1cbca939dd36913c143aead8f91947eb0cb56.

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Heiberg 1881 volume title. Complete source page. PDF page 12. SHA-256 66870c9d59cdf76566f33e51db4c3df01698355b8b8c7203a79c802bd41a8b87.

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